

Less Token, More Signal [A Reproduction]

What the released STEP code independently supports—and what it does not.

Independent reproduction ⓘ · ICML 2026 paper 21218 · OpenReview 4iupzej9nT

1 Scope before scores

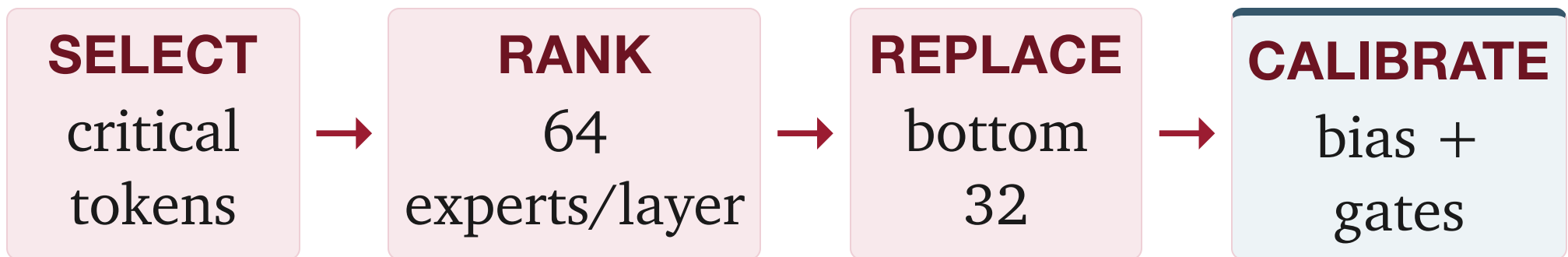
The paper claims results on **Qwen3-30B-A3B**. The released repository only includes runnable OLMoE support; Qwen imports and branches are commented and its model implementation is absent.

- Faithful released proxy: OLMoE-1B-active/7B-total.
- Exact 50% expert sparsity and full 128-step calibration schedule.
- One A100 80 GB; immutable Job and raw JSON retained.

Rule: proxy evidence is never relabelled as Qwen3-30B replication.

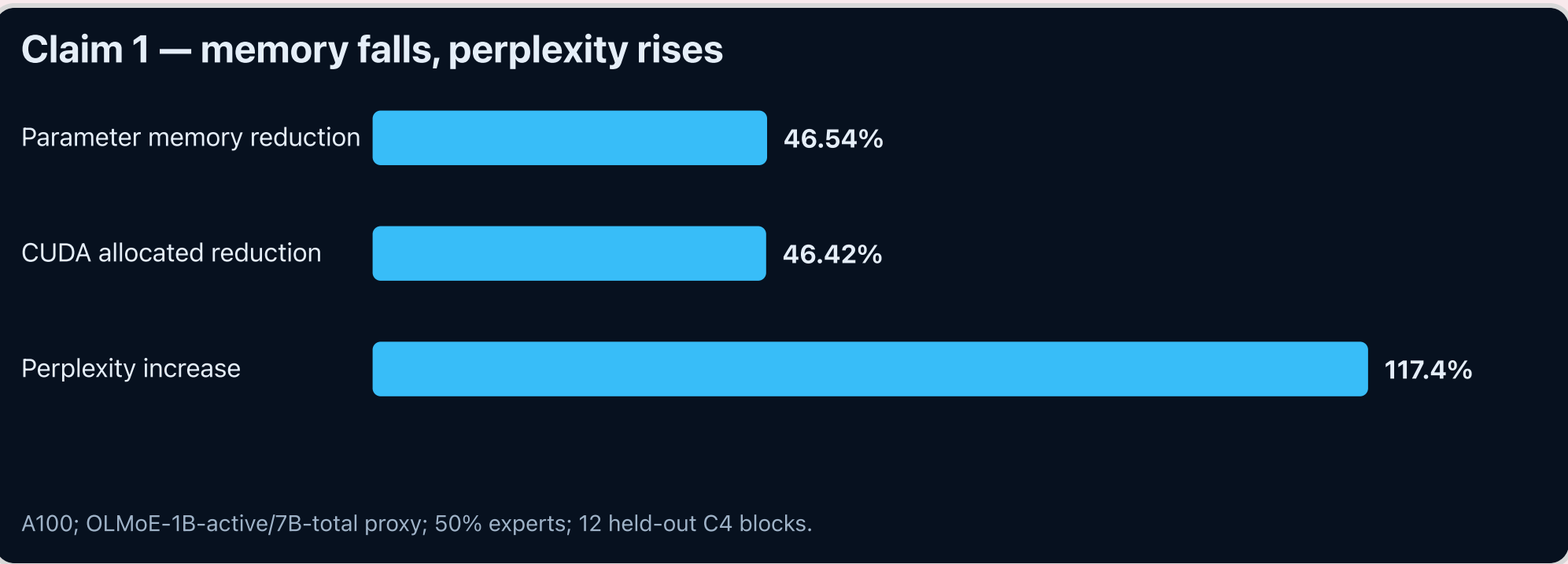
2 Released STEP procedure

Attention selects critical token positions; routing and expert-output statistics rank experts; removed experts become lightweight learned biases; a short calibration pass updates biases and gates.



Revision: 03fdea9a . Settings: layer-wise fusion ranking, tau 0.5, C4 calibration, using BF16 arithmetic.

3 Claim 1: memory vs quality



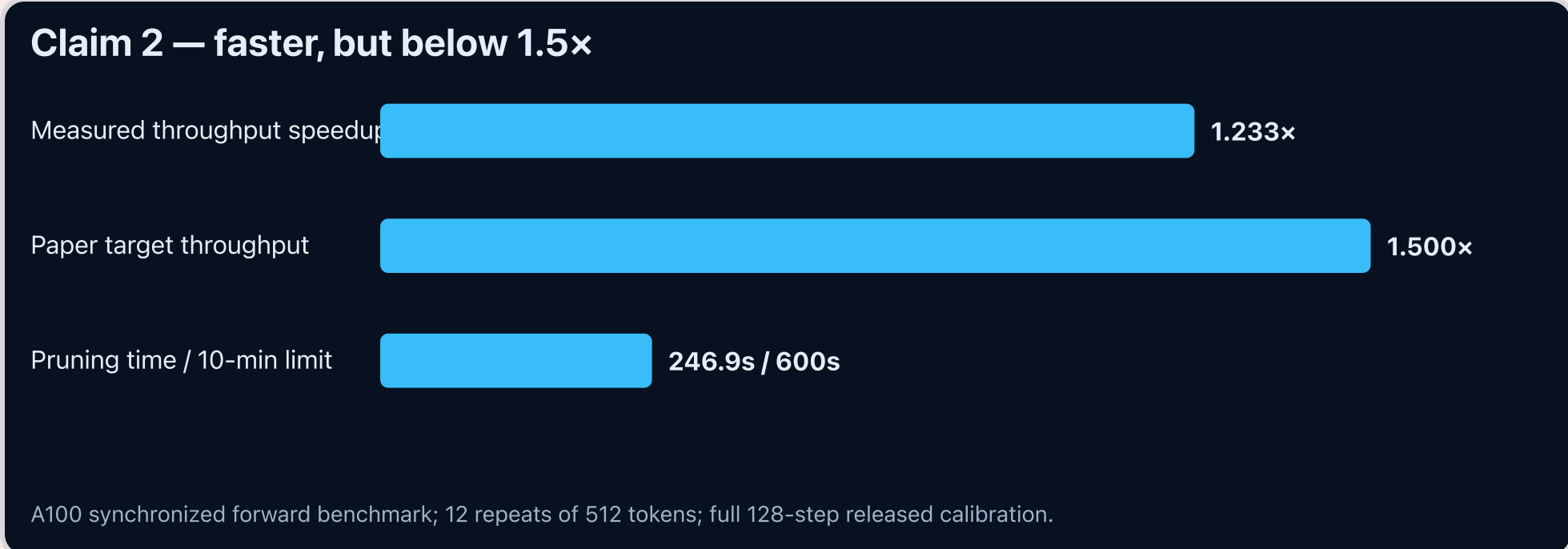
New measurement. Exactly 512 of 1,024 layer-experts were replaced.

46.54% parameter bytes reduced **46.42%** CUDA allocation reduced **2.17×** perplexity ratio, worse

PARTIAL/PROXY: the near-50% memory direction reproduced. “Minimal degradation” did not: paired loss delta +0.777, bootstrap 95% CI [0.555, 1.012].

Mixed: memory yes; quality no.

4 Claim 2: speed + runtime



New measurement. Synchronized A100 forwards; 12 repeats of 512 tokens.

1.233× measured throughput **1.500×** paper target **246.9 s** pruning procedure

PARTIAL/PROXY: under 10 minutes reproduced. The 1.233× speedup (95% CI 1.225×–1.248×) missed the paper target.

5 Reproducible evidence packet

- Compute:** A100 80 GB; PyTorch 2.3.1; Transformers 4.55.3.
- Data:** 262,144 calibration tokens; 6,144 validation tokens.
- Provenance:** code, stdout, timings, losses, Job URL, and Bucket JSON.
- Cost:** \$0.24 for 347 seconds.

Job logs and Bucket JSON agree.

All headline numbers come from new raw measurements.

6 Component scorecard

Memory	Supported
Quality	Not supported
Throughput	Not supported
Prune time	Supported

Each compound verdict is partial because one essential component failed.

7 Takeaways ★

- Memory:** near-50% reduction transfers to the released proxy.
- Quality:** perplexity more than doubled.
- Throughput:** 1.233× vs 1.5× claimed.
- Pruning:** 246.9 seconds, comfortably under 10 minutes.
- Scientific limit:** missing Qwen release prevents direct 30B verification.